SIGNAL INTELLIGENCE DOCUMENT

RigaCap

Signal Intelligence: How RigaCap Reads the Market

A Complete Technical Guide to Our Trading Signal Pipeline — From Raw Data to Portfolio Execution

4,000+	7	5 min	12%
STOCKS SCANNED	MARKET REGIMES	INTRADAY MONITORING	TRAILING STOP

Confidential — April 2026

v2.0.0 · 3-Factor Ensemble · 7-Regime Intelligence

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Executive Summary

RigaCap's signal intelligence pipeline transforms raw market data from 4,000+ US equities into high-conviction trading signals through a three-pillar approach:

DWAP Timing

Proprietary breakout detection from 200-day volume-weighted accumulation zones.

Momentum Quality

Multi-timeframe momentum ranking with volatility-adjusted composite scoring.

Regime Intelligence

7-state market classifier that adapts strategy parameters to current conditions.

Every signal must pass all three filters before entry. Once invested, positions are monitored every 5 minutes during market hours with adaptive trailing stops. The entire pipeline has been validated through 10-year walk-forward testing with zero hindsight bias, producing **+603% cumulative return over 10 years** (~33% annualized, 1.19 Sharpe, 30% max drawdown).

1. Data Acquisition Pipeline

The pipeline begins with a comprehensive stock universe spanning NASDAQ and NYSE exchanges. Every trading day, the system fetches, filters, and enriches price data to feed the signal engine.

Stock Universe

PARAMETER	VALUE	SOURCE
Total Universe	~4,000 symbols	NASDAQ + NYSE exchanges
Minimum Price	$15	`config.MIN_PRICE`
Minimum Volume	500,000 shares/day	`config.MIN_VOLUME`
Lookback Period	7 years OHLCV	`fetch_data(period="7y")`
Data Source	Alpaca SIP (primary) + yfinance (fallback)	Split/dividend adjusted, consolidated feed

Exclusion Rules

The following instrument types are automatically excluded from the universe:

- **Leveraged ETFs:** TQQQ, SQQQ, SSO, SDS, QLD, QID, TNA, TZA
- **Inverse Products:** SPXU, UVXY, SVXY
- **Volatility Products:** VXX
- **Sector-leveraged:** FAS, FAZ

Fetch Architecture

PARAMETER	VALUE	PURPOSE
Batch Size	100 symbols (Alpaca) / 10 (yfinance)	Alpaca supports larger batches
Max Retries	2 per batch	Handle transient failures
Batch Delay	1.5 seconds	Rate limit compliance
Retry Delay	3.0 seconds (2× batch delay)	Backoff on failure

PARAMETER	VALUE	PURPOSE
Required Symbols	SPY, ^VIX	Always fetched for regime detection

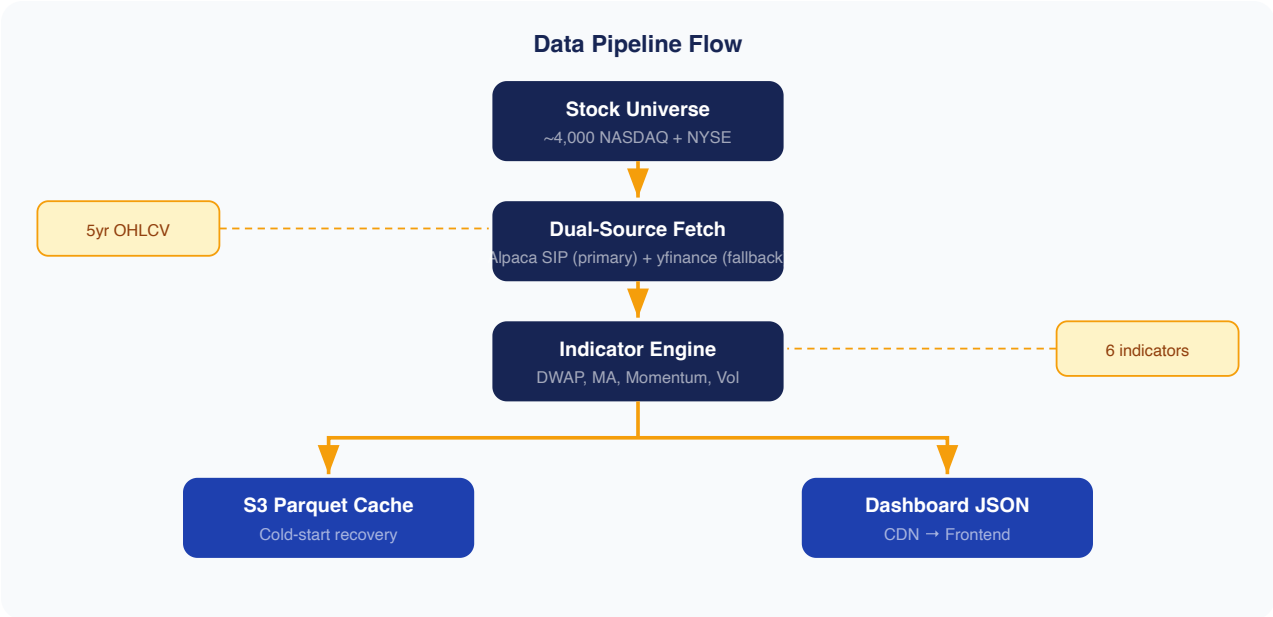
After the initial 5-year fetch, subsequent updates use **incremental mode** — only downloading prices since the last cached date. This reduces daily update time from minutes to seconds. All price data is persisted to S3 as a consolidated gzipped pickle for Lambda cold-start recovery, with a parallel Parquet shadow write (columnar, typed) enabling ad-hoc SQL queries via DuckDB.

Data Integrity Guardrails (added Apr 2026)

Split-adjustment and data-corruption handling are now built into every layer of the pipeline:

- **Split-adjusted by default:** the Alpaca `StockBarsRequest` always uses `Adjustment.SPLIT`, preventing the class of bug where an unadjusted stock split (NVDA 10:1, GOOG 20:1, etc.) creates an artifact -90% "crash" in the historical series.
- **Abrupt-jump rejection:** any single-day close-to-close change greater than 65% (log-return > 0.5) on a symbol's recent history marks it as corrupted. Real market moves rarely exceed this; unadjusted corporate actions always do.
- **DWAP-ratio sanity:** symbols where the current close is > 50× or < 0.02× the 200-day DWAP are rejected as clearly corrupted (ticker reuse or missed adjustment).
- **Long-gap rejection:** any symbol with a >10-day gap between consecutive trading days is flagged as ticker-reuse corruption (old entity delisted, ticker recycled to a new entity).
- **Nightly Layer 2 hygiene:** Alpaca's corporate-actions API is polled every weekday at 6:30 PM ET. Splits trigger automatic force-refetch with `Adjustment.SPLIT`. Ticker reuses trigger auto-quarantine. Admin digest email reports status.
- **Asset-ID tracking:** every symbol's Alpaca UUID is stored in `symbol_metadata`. When an exchange recycles a ticker to a new entity, the UUID changes, and the system detects the swap before the next signal fires.

Before these guardrails, undetected splits inflated backtest returns and could trigger false sell alerts on live positions (entry price stored at pre-split basis, next scan evaluates against post-split price → fake -90% loss → trailing stop triggers). The pipeline now handles splits end-to-end without manual intervention.

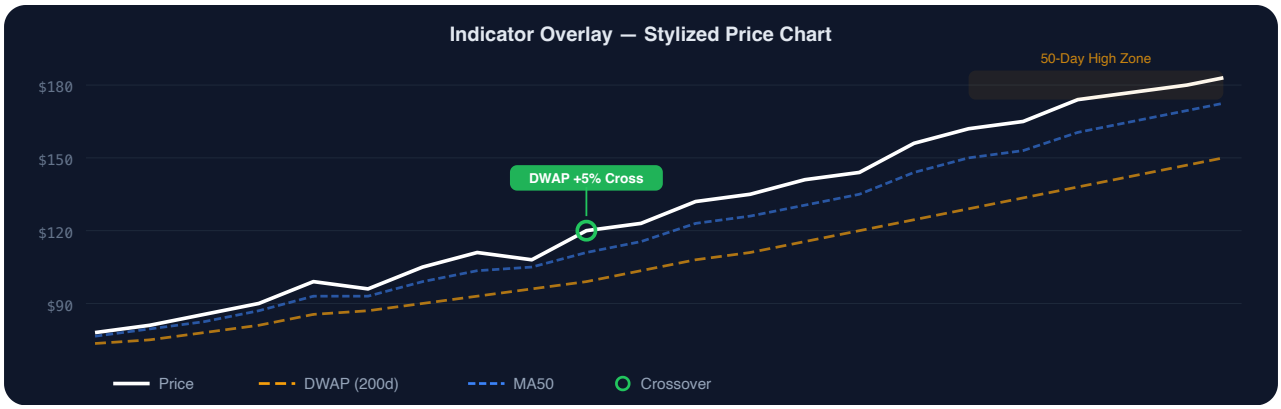

Data Pipeline Flow
Stock Universe
~4,000 NASDAQ + NYSE
5yr OHLCV
Dual-Source Fetch
Alpaca SIP (primary) + yfinance (fallback)
Indicator Engine
DWAP, MA, Momentum, Vol
6 indicators
S3 Parquet Cache
Cold-start recovery
Dashboard JSON
CDN → Frontend

2. Technical Indicator Engine

Six core indicators are computed for every stock in the universe. These indicators feed both the DWAP timing signal and the momentum quality ranking.

INDICATOR	FORMULA	PERIOD	PURPOSE
DWAP	`Σ(price × volume) / Σ(volume)`	200 days	Volume-weighted accumulation zone
MA20, MA50, MA200	Simple Moving Average	20 / 50 / 200 days	Trend confirmation, quality filter
Short Momentum	`(price / price[t−10] − 1) × 100`	10 days	Short-term price acceleration
Long Momentum	`(price / price[t−60] − 1) × 100`	60 days	Sustained trend strength
Volatility	`std(returns, 20) × √252 × 100`	20 days	Risk-adjusted scoring penalty
Distance from High	`(price / max(price, 50d) − 1) × 100`	50 days	Breakout proximity detection

> **Lazy Computation:** Indicators are computed on-demand and cached in the `data_cache` dictionary. The first scan computes all indicators; subsequent scans reuse cached values unless data is refreshed.



3. The Ensemble Signal — Three-Factor Entry

A stock must satisfy all three independent factors simultaneously before the system generates a buy signal. This multi-confirmation approach dramatically reduces false signals and ensures only the highest-conviction setups enter the portfolio.

Factor 1: Breakout Detection (Timing)

A proprietary breakout indicator detects when a stock clears its long-term accumulation zone with conviction. The breakout sensitivity threshold is **dynamically optimized** every two weeks by Bayesian optimization (TPE), adapting to current market conditions. Typical range: 3–7% above the baseline.

Factor 2: Momentum Quality (Ranking)

Stocks passing the breakout filter are ranked by a composite momentum score combining short-term acceleration, long-term trend strength, and a volatility penalty. All three weights and lookback windows are **adaptively tuned** by the optimizer every two weeks. Typical lookback: 5–20 days (short), 40–90 days (long).

Factor 3: Confirmation Filters

FILTER	CONDITION	PURPOSE
Breakout Proximity	Near recent highs (adaptive, 2–8%)	High conviction entries only
Volume	≥ 500,000 shares/day	Sufficient liquidity
Price	≥ $15	Filters penny stocks
Position Sizing	4–8 positions at 12–20% each (adaptive)	Concentration adapts to regime

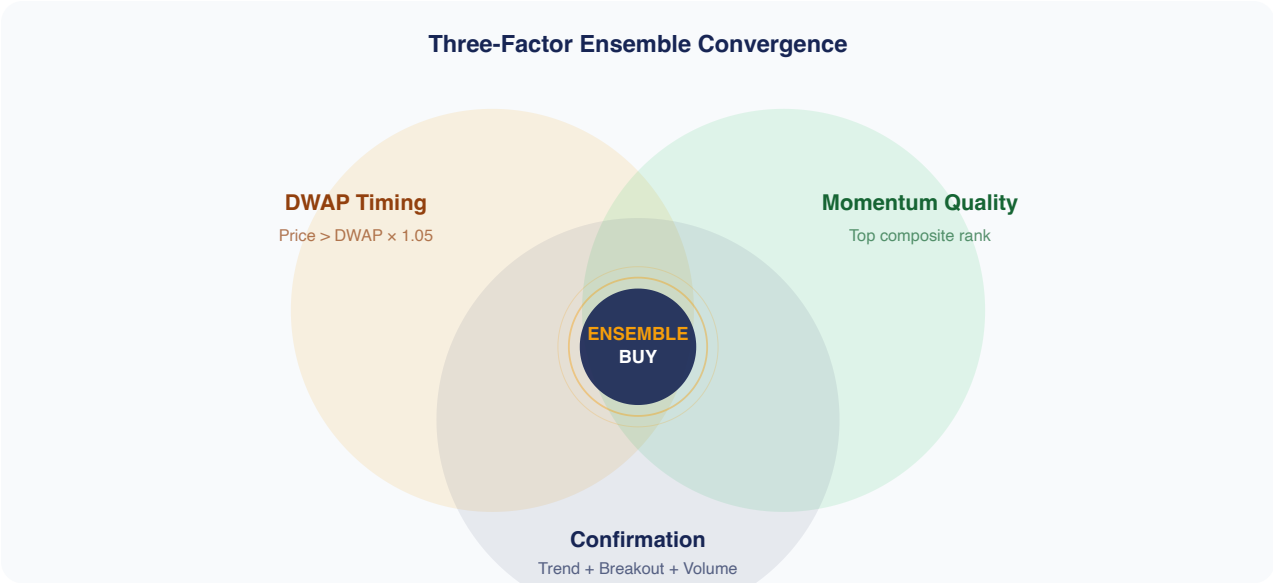
Adaptive Optimization: Every two weeks, the system re-optimizes 22 strategy parameters using 100 Bayesian trials against the most recent 60 days of market data. Parameters include breakout sensitivity, momentum weights, position concentration, trailing stop width, and profit-lock thresholds. No parameter is fixed — the system adapts to what works in the current regime.

Anti-Squeeze Filters

A fourth layer of rejection criteria was added after a Feb 2021 forensic review revealed the strategy had entered meme-stock squeezes at their peaks (EH, TLRY, LCID) and taken catastrophic losses. These filters reject candidates that have already run parabolically before our other factors fire:

FILTER	DEFAULT / RANGE	PURPOSE
Max Recent Return	`max_recent_return_pct` : 30–1000 (tuned by TPE)	Reject candidates up > X% in the last 30 days — prevents chasing tops
Price Velocity Cap	`price_velocity_cap_pct` : 15–1000 (tuned by TPE)	Reject candidates up > X% in the last 5 days — catches parabolic moves
RSI Overbought	`rsi_oversold_filter` : 60–100 (100 = disabled)	Reject candidates above RSI threshold — standard overbought guard

Each parameter is tuned by the TPE optimizer. Value 1000 effectively disables the filter, letting Bayesian optimization learn whether the constraint helps or hurts for a given regime.



4. Market Regime Intelligence

The market regime classifier analyzes SPY price action, VIX levels, market breadth, and new-highs data to classify the current environment into one of 7 distinct regimes. Strategy parameters automatically adapt to each regime.

7 Regime Classifications

Strong Bull
SPY +5% above 200MA
VIX < 40th percentile

Weak Bull
SPY 0–10% above 200MA
Moderate volatility

Rotating Bull
SPY choppy, 1yr gain +5%
Sector rotation

Range-Bound
SPY ±5% of 200MA
Flat 1-year return

Weak Bear
SPY −15 to 0% vs 200MA
Elevated VIX

Panic / Crash
SPY < −10% vs 200MA
VIX > 70th percentile

Recovery
Bouncing from lows
SPY 0–15% above 50MA

+ Forecasting
1–2 week transition probability outlook

Scoring Weights

Regime detection uses weighted scoring across condition indicators. Higher weights mean that indicator is more influential in regime classification:

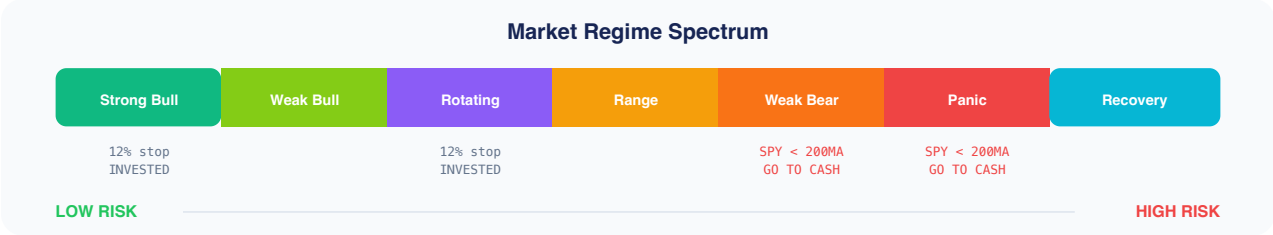
INDICATOR	WEIGHT	ROLE
Long-term Trend (252d SPY return)	2.5×	Key differentiator: rotating bull vs range-bound
SPY vs 200MA	1.5×	Primary bull/bear boundary
New Highs %	1.5×	Breadth confirmation of trend
Breadth % (above 50MA)	1.2×	Market participation width
VIX Percentile, Trend Strength, SPY vs 50MA	1.0×	Standard conditions

Regime Response

The system detects 7 market regimes and responds at two levels: (1) a **binary market filter** that moves to cash when SPY drops below its 200-day moving average, and (2) **adaptive parameter optimization** that adjusts position sizing, trailing stops, and entry sensitivity based on what the optimizer learned in the most recent 60-day window.

REGIME	TRAILING STOP	MAX POSITIONS	POSITION SIZE	ACTION
Strong Bull	~11%	~6	~18%	Stay invested, spread bets
Rotating Bull	~14%	~6	~17%	Stay invested, fast momentum
Range-Bound	~14%	~5	~20%	Selective, concentrated
Weak Bear	~14%	~4	~20%	Fewer, bigger conviction bets
Panic/Crash	Cash — SPY below 200-day MA			Go to cash
Recovery	~16%	~6	~14%	Cautious re-entry, wide stops

Values shown are averages from the optimizer's choices across 138 periods. Actual parameters are re-optimized every two weeks and may differ from these regime averages.



Transition Forecasting

Beyond static classification, the regime engine computes **indicator trajectories** to predict where the market is heading in the next 1–2 weeks:

- **VIX Delta (5-day):** Rising VIX boosts bearish regime probabilities
- **Trend Acceleration:** Compares recent 5-day return vs prior 5-day return
- **200MA Proximity:** Within 2% of 200MA triggers higher transition uncertainty
- **Sharp Moves:** 5-day returns exceeding ±5% boost crash/recovery probabilities

The forecast produces a **recommended action** ranging from `stay_invested` (SPY > 200MA) → `go_to_cash` (SPY < 200MA). Binary action — no intermediate steps.

5. Intraday Monitoring (Every 5 Minutes)

Once positions are entered, the intraday monitor checks live prices every 5 minutes during market hours. This enables rapid response to trailing stop hits and regime-driven exits.

Schedule

PARAMETER	VALUE
Hours	Monday–Friday, 9:00–15:55 ET
Interval	Every 5 minutes
Price Source	Alpaca latest quote (SIP) / yfinance fallback
Alert Deduplication	`(position_id, action, date)`

Trailing Stop Mechanics

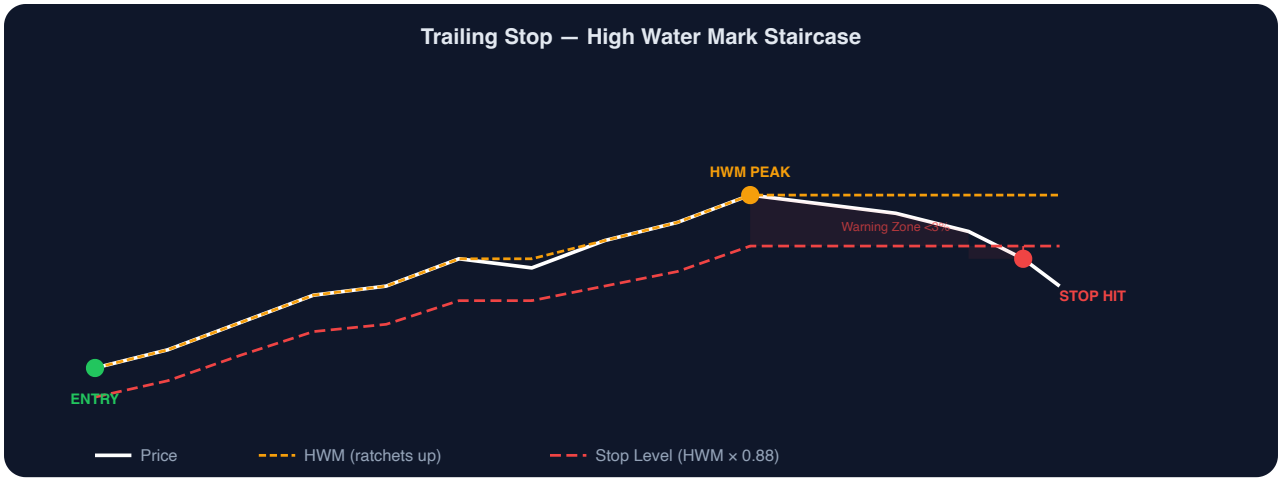
The trailing stop is calculated from the **high water mark** (HWM) — the highest price since entry, not the entry price itself:

```
HWM = max(entry_price, stored_highest, current_price)
stop_level = HWM × (1 − 0.12)

SELL when price ≤ stop_level     WARNING when distance < 3%
```

Exit Priority

1. **Regime Exit (go_to_cash):** Immediate sell regardless of P&L — capital preservation
2. **Trailing Stop Hit:** Price ≤ HWM × 0.88 — overrides regime warnings
3. **Warning Zone:** Within 3% of stop level — email alert, no action
4. **Regime Warning:** SPY approaching 200MA — email alert



Intraday DWAP Crossover Detection

The monitor also watches stocks on the **dashboard watchlist** (stocks approaching the DWAP threshold). If a watchlist stock crosses DWAP +5% intraday, the system:

- Adds the stock to the dashboard as a fresh intraday signal
- Removes it from the watchlist
- Sends intraday signal alert emails to subscribed users (skipping users who already hold the stock)
- Updates the S3 dashboard JSON in real-time

6. Position Management & Sizing

The model portfolio service maintains two parallel portfolios — live (intraday monitoring) and walk-forward (biweekly rebalancing) — plus three ghost portfolios for comparison.

Portfolio Parameters

6	15%	12%	14d
MAX POSITIONS	POSITION SIZE	TRAILING STOP	REBALANCE PERIOD

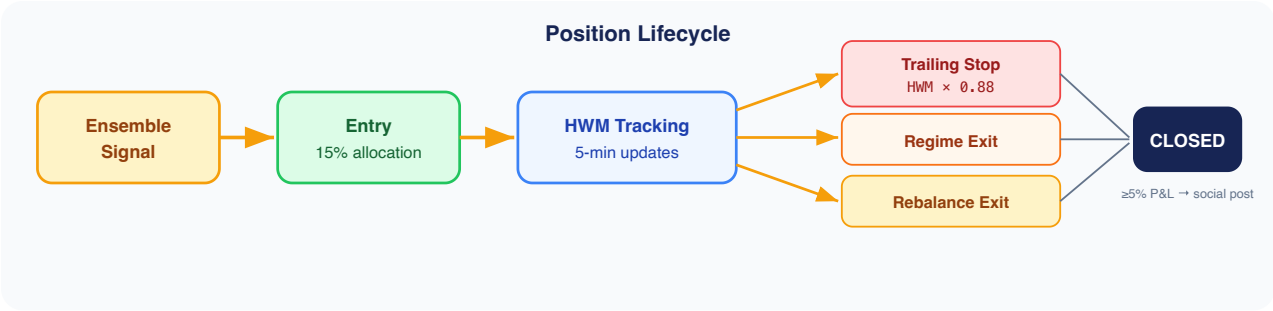
Entry Logic

- Read fresh ensemble signals from dashboard cache (S3 JSON)
- Filter to `is_fresh = true` only, skip symbols already held
- Sort by ensemble_score (highest first)
- Enter up to `MAX_POSITIONS - open_count` new positions
- Allocate 15% of **current cash** per position (not starting capital)
- Minimum allocation: $100

Exit Paths

EXIT REASON	TRIGGER	PORTFOLIO
`trailing_stop`	Price ≤ HWM × 0.88	Both live & WF
`regime_exit`	`go_to_cash` recommendation	Live only
`rebalance_exit`	Biweekly period boundary	Walk-forward only

> **Biweekly Boundaries:** Walk-forward periods are 14-day intervals anchored to February 1, 2026 (`WF_ANCHOR_DATE`). At each boundary, all open WF positions are force-closed and fresh signals are entered for the new period.



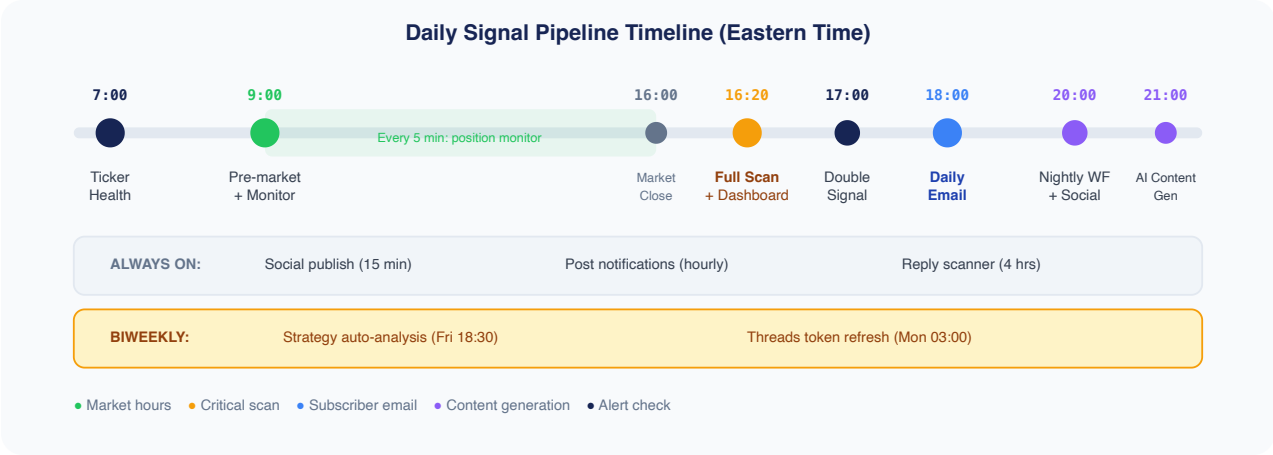
Ghost Portfolios (Parallel Universes)

Three alternative portfolios run alongside with different parameters to prove the chosen configuration:

GHOST	TRAILING STOP	MAX POSITIONS	POSITION SIZE	STYLE
Aggressive	8%	8	12%	Tight stops, more positions
Conservative	18%	4	20%	Wide stops, fewer positions
Top-3 Only	12%	3	30%	Concentrated best picks

7. Alert & Communication Pipeline

A coordinated set of scheduled jobs runs throughout each trading day, from pre-market data refresh through nightly content generation. Every job is managed by EventBridge with US Eastern timezone awareness.



Alert Types

ALERT	TRIGGER	AUDIENCE	CHANNEL
Daily Digest	6 PM ET, trading days	All subscribers	Email + Push
Sell Alert	Trailing stop or regime exit	Position holders	Email + Push
Warning Alert	Within 3% of stop level	Position holders	Email + Push
Double Signal	DWAP + momentum crossover within 3 days	All subscribers	Email
Intraday Signal	Watchlist stock crosses DWAP +5%	Subscribers (not holding)	Email
Ticker Health	Data issues detected at 7 AM	Admin only	Email

Email Deliverability: All emails include SPF + DKIM + DMARC authentication, RFC 8058 List-Unsubscribe headers for one-click unsubscription, and per-category preference controls.

8. Walk-Forward Validation

Walk-forward testing is the gold standard for trading system validation. Unlike traditional backtesting which optimizes on all available data (introducing hindsight bias), walk-forward divides history into sequential periods and tests forward into data the optimizer has never seen.

Methodology

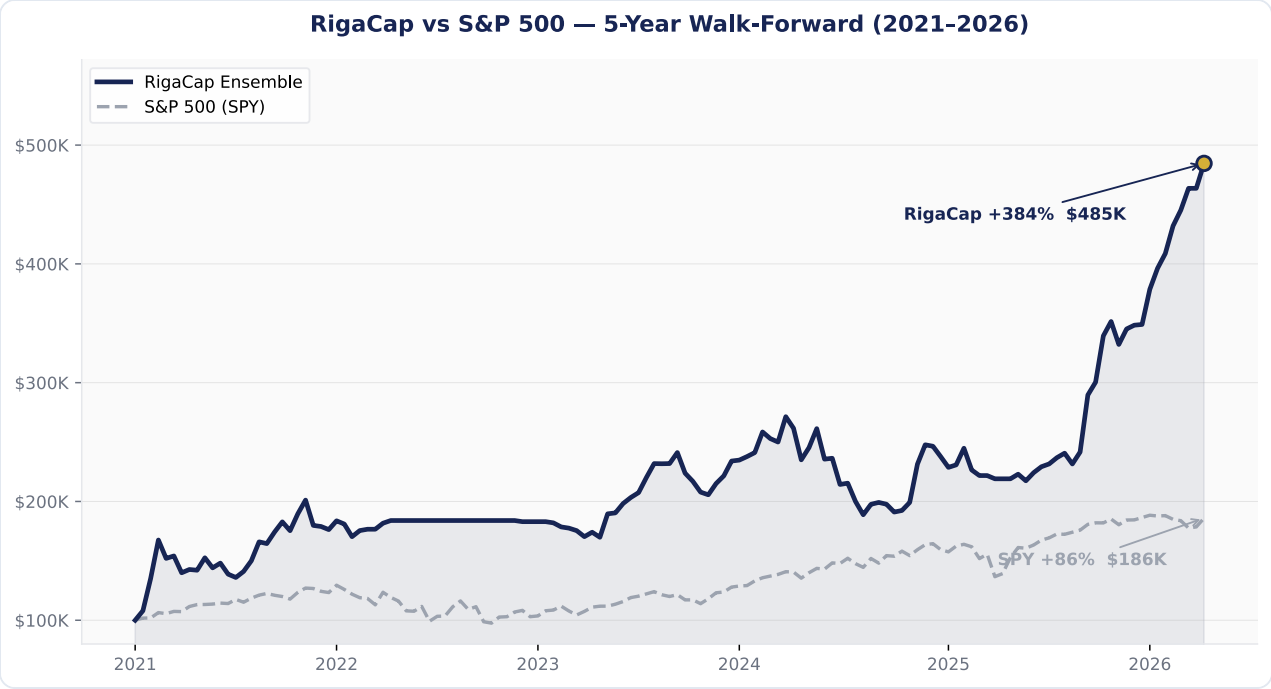
- **Period Length:** Biweekly (14-day intervals, anchored Feb 1)
- **Optimization:** Uses only historical data up to each period boundary
- **Trading:** Enters fresh ensemble signals, force-closes at period end (`rebalance_exit`)
- **Position Carry:** Carry-over outperforms force-close (A/B tested: +206% vs +121%)
- **Max Symbols:** 100 per period for computational efficiency
- **Strategy:** Ensemble (strategy_id: 5) — DWAP + Momentum + Regime

> **Zero Hindsight Bias:** At each biweekly boundary, the system knows nothing about what happens next. Parameters are optimized on past data, then the system trades forward into unseen market conditions. The +603% return (10yr) is entirely out-of-sample.

Walk-Forward Performance Summary

WINDOW	RIGACAP	S&P 500	RISK-ADJUSTED
5-Year Avg	**+384%**	**+84%**	**~33% annualized, Sharpe 1.19, 30% max drawdown**
10-Year Total	**+603%**	**+257%**	**~33% annualized, 30% max drawdown**

5-Year Avg averages the Trial 37 walk-forward across 8 different start dates — no cherry-picking.



RigaCap Ensemble (solid) vs S&P 500 (dashed) — Feb 2021 to Feb 2026 with market regime overlays

RigaCap vs. Benchmarks (5-Year Cumulative)

RigaCap	+603% (10yr)
S&P 500	+257% (10yr)

Key Insights

2022 Resilience

When the S&P 500 fell -20% in 2022, RigaCap stayed flat. The 7-regime detection moved to cash before the worst of the drawdown.

Controlled Drawdowns

Max drawdown of 30% across the 5-year walk-forward — the 14% trailing stop, SPY regime exit, and Cascade Guard prevented catastrophic losses.

Cascade Guard

Portfolio-Level Circuit Breaker

When 3+ positions hit trailing stop on the same day, the system pauses all new entries for 10 days. Prevents the “buying into a falling knife” pattern where replacement positions also fail. Triggered 7 times in 5.3 years — the system is fully active 95% of the time.

+87pp Return Impact

Grid-searched across 12 configurations. This single guardrail lifted returns from +297% to +384% by avoiding concentrated re-entry losses during cascade events, without altering signal quality or reducing time in the market.

How We Trade

Winners average +15%

Average winner: +15.2%. Win/loss ratio: 1.77x — winners are nearly twice the size of losers. The trailing stop lets profits compound rather than capping them.

Losers cut at −9%

Average loser: -8.6%. 48.6% win rate — you don’t need to win most trades when your winners are 1.77x your losers. The system admits fast when a trade isn’t working.

Stats from walk-forward simulation 1032 (333 trades over 5.3 years, adaptive + Cascade Guard).

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